UMGS: UAV Multispectral Gaussian Splatting for Spectral-Index Synthesis

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SUBMISSION ID: TBD

UAV multispectral reconstruction is difficult not simply because more spectral channels must be rendered, but because non-RGB bands are weak geometry drivers. G/R/RE/NIR images often contain lower texture, different appearance statistics, and less reliable cross-spectral correspondences than RGB, while practical UAV rigs introduce cross-FOV, resolution, and alignment differences between RGB and multispectral cameras. Recent multispectral and hyperspectral Gaussian methods demonstrate that spectral rendering and vegetation-index synthesis are feasible, but many rely on specialized neural spectral representations, registered inputs, or custom rendering and evaluation paths. This leaves a practical gap: reconstructing raw UAV multispectral captures into stable, product-ready Gaussian splats that preserve spectral fidelity while remaining compatible with standard 3DGS-style rendering and evaluation workflows.

We propose UMGS, a two-stage UAV multispectral Gaussian splatting framework for spectral-index synthesis. Stage 1 learns an RGB anchor with standard Gaussian splatting. Stage 2 transfers spectral appearance while freezing geometry, opacity, and Gaussian identity. We study two locked variants: UMGS-I, independent per-band transfer from the shared RGB anchor, and UMGS-J, joint band-bank transfer in one unified multispectral state with a single authoritative checkpoint and per-band exported evaluation views.

We evaluate on a 10-scene UAV-focused benchmark, including 7 raw UAV captures and 3 aligned aerial multispectral scenes. UMGS-I and UMGS-J are numerically near-tied on this benchmark and show matched relative-depth behavior under held-out probe views. On aligned aerial scenes, common-mask comparison against MS-Splatting is competitive and scene-dependent. Ablations confirm the role of geometry locking: Scratch-MS and Unfrozen-support can improve selected single-band metrics, but they break the shared Gaussian support needed for model-level multispectral products and index synthesis.

Overall, the results support geometry locking as a practical design for UAV multispectral Gaussian splatting and stable spectral-index synthesis.

CCS Concepts: • **Computing methodologies** → **Image-based rendering**; *Reconstruction*; • **Applied computing** → *Earth and atmospheric sciences.*

Additional Key Words and Phrases: multispectral imaging, gaussian splatting, 3D reconstruction, UAV

ACM Reference Format:
Anonymous Author(s). 2026. UMGS: UAV Multispectral Gaussian Splatting for Spectral-Index Synthesis. 1, 1 (May 2026), 6 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

UAV multispectral imaging is increasingly used in precision agriculture and environmental monitoring, where decisions are made from cross-band products rather than from a single rendered view [Colomina and Molina 2014; Mathews and Jensen 2013; Turner et al. 2015; Zhang and Kovacs 2012]. In this setting, reconstruction quality has two coupled requirements: (1) high per-band fidelity and (2) stable cross-band geometric correspondence. The second requirement is critical for index synthesis (e.g., NDVI, GNDVI, NDRE) because even small support drift between bands can bias ratio-based measurements [Gitelson et al. 1996; Goetz et al. 1985; Huete 1988; Manolakis et al. 2016; Rouse et al. 1974].

Raw UAV multispectral data make this requirement difficult to satisfy. Non-RGB bands typically provide weaker texture and less stable local features than RGB, so direct structure-from-motion or Gaussian optimization from a single spectral band is often underconstrained. At the same time, practical multispectral rigs capture RGB and G/R/RE/NIR through different cameras with different fields of view, resolutions, and spectral responses. A reconstruction method must therefore handle both weak non-RGB geometry cues and cross-camera alignment before it can produce reliable model-level products.

Recent radiance-field and Gaussian methods greatly improved training speed and view quality [Barron et al. 2022; Huang et al. 2024; Kerbl et al. 2023; Mildenhall et al. 2020; M"uller et al. 2022; Yu et al. 2024]. Emerging multispectral, hyperspectral, and thermal Gaussian lines [Chen et al. 2024; Lu et al. 2024a; Meyer et al. 2025; Thirgood et al. 2025] further show that spectral rendering and index synthesis are feasible. Our focus is a narrower deployment problem: converting raw UAV multispectral captures into product-ready Gaussian splats while avoiding unstable non-RGB geometry optimization, cross-band misalignment, and dependence on custom spectral renderers or evaluation paths.

We introduce **UMGS**, a UAV multispectral Gaussian splatting framework for spectral-index synthesis. UMGS formulates multispectral reconstruction as *RGB-anchored geometry locking*: geometry, opacity, and Gaussian identity are learned once from RGB and then frozen during G/R/RE/NIR transfer. This design turns RGB into a stable geometric carrier, while each spectral band only learns appearance on the same splat support. We evaluate two implementations of the same constraint:

- **UMGS-I**: independent per-band transfer from a shared RGB anchor.
- **UMGS-J**: joint multispectral training with band-specific appearance banks in one unified checkpoint.

Both preserve identical support by construction and export standard 3DGS-style per-band evaluation views; they differ only in how spectral appearance parameters are organized and optimized.

For raw UAV captures, alignment quality is another bottleneck. Cross-band FOV and resolution mismatch can make naive baselines unreliable, especially for NIR. We therefore use a baseline-aware

ACM XXXX-XXXX/2026/5-ART
https://doi.org/10.1145/nnnnnnn.nnnnnnn

rectification QA protocol as supporting infrastructure: it separates strict pass, pass-with-warning, and fail, and logs per-frame matching/stability diagnostics rather than relying on a single opaque gate.

Our main evaluation covers a 10-scene UAV-focused benchmark: 7 raw UAV captures and 3 aligned aerial multispectral scenes. The raw track validates end-to-end applicability (alignment, QA, reconstruction, products, and geometry diagnostics). The aligned track removes rectification confounds and supports fair common-mask comparison with MS-Splatting. Across this suite, UMGS-I and UMGS-J are near-tied and consistently preserve shared support. Scratch-MS and Unfrozen-support are useful upper-bound ablations for single-band fitting, but they do not satisfy the product-level topology contract required for stable multispectral index synthesis.

The contributions are:

- A geometry-locked multispectral Gaussian formulation that uses RGB as the geometric carrier for product-ready spectral-index synthesis.
- Two concrete UMGS realizations implementing the same lock with independent (UMGS-I) and joint (UMGS-J) spectral parameterizations.
- A UAV-focused benchmark protocol with image, index, depth-reference, cost, and external common-mask comparisons under unified reporting.
- Supporting raw-scene infrastructure for baseline-aware rectification QA with explicit diagnostic states and decision inputs.

2 Related Work

2.1 Radiance Fields and Gaussian Scene Representations

NeRF established neural volumetric rendering as a practical paradigm for novel view synthesis [Mildenhall et al. 2019, 2020]. Follow-up work improved anti-aliasing, unbounded scenes, and training speed, including mip-NeRF/mip-NeRF360, Instant-NGP, Plenoxels, TensoRF, and DVGO [Barron et al. 2022, 2021; Chen et al. 2022; Fridovich-Keil et al. 2022; M"uller et al. 2022; Sun et al. 2022]. Large-scene and explicit-factorized lines such as NSVF, K-Planes, Zip-NeRF, MERF, Mega-NeRF, and Block-NeRF further explored memory/scale trade-offs [Barron et al. 2023; Fridovich-Keil et al. 2023; Liu et al. 2020; Reiser et al. 2023; Tancik et al. 2022; Turki et al. 2022].

3D Gaussian Splatting (3DGS) moved from volumetric integration to explicit anisotropic splats, enabling faster optimization and real-time rendering [Kerbl et al. 2023]. Subsequent work improved alias control, structural priors, compactness, and scene scale, including Mip-Splatting, Scaffold-GS, 2DGS, SuGaR, LightGaussian, Compact3DGS, Dynamic 3DGS, CityGaussian, and CityGaussianV2 [Fan et al. 2024; Guédon and Lepetit 2024; Huang et al. 2024; Lee et al. 2024; Liu et al. 2024, 2025; Lu et al. 2024b; Luiten et al. 2024; Yu et al. 2024]. These methods target rendering quality, memory, or geometry fidelity in RGB-dominant settings. UMGS keeps the standard 3DGS-style representation useful for UAV deployment, but changes how spectral appearance is attached to the geometry.

2.2 Multispectral and Thermal Gaussian Reconstruction

Multispectral remote sensing has long emphasized calibration, cross-band consistency, and physically meaningful indices [Gitelson et al. 1996; Goetz et al. 1985; Huete 1988; Manolakis et al. 2016; Rouse et al. 1974]. In UAV workflows, geometric preprocessing and modality alignment are often the limiting factor for downstream reliability [Colomina and Molina 2014; Mathews and Jensen 2013; Turner et al. 2015; Zhang and Kovacs 2012]. While deep learning for spectral imagery is mature in classification/segmentation tasks [Audebert et al. 2019; Li et al. 2019; Paoletti et al. 2019; Zhong et al. 2018], 3D multispectral scene reconstruction is still emerging.

Recent Gaussian-based spectral works include MS-Splatting for multi-camera multispectral NVS, HyperGS for hyperspectral latent rendering, MSGS for wavelength-aware Gaussian modeling, and thermal-domain lines such as ThermalGaussian and Thermal3D-GS [Chen et al. 2024; Lu et al. 2024a; Meyer et al. 2025; Thirgood et al. 2025; Zhang et al. 2026]. These works confirm the feasibility of spectral Gaussian rendering and, in several cases, already store multiple spectral or modal signals in one Gaussian-based representation. MS-Splatting, for example, learns compact per-splat features and decodes spectral values with an MLP; HyperGS performs hyperspectral rendering in a learned latent space; ThermalGaussian learns registered RGB/thermal Gaussians after camera calibration. Our focus is complementary: raw UAV multispectral deployment, where non-RGB bands are weak geometry drivers, RGB/MS cameras can differ in FOV and resolution, and downstream products benefit from standard 3DGS-style per-band artifacts rather than only a custom neural spectral decoder.

2.3 SfM, Matching, and Cross-Modal Alignment

Practical reconstruction still depends on robust SfM/MVS foundations [Fischler and Bolles 1981; Hartley and Zisserman 2004; Sch"onberger and Frahm 2016; Sch"onberger et al. 2016]. Handcrafted and learned feature matching both remain relevant, from SIFT/SURF/ORB and ECC-style alignment to SuperPoint, SuperGlue, LoFTR, LightGlue, and RoMa [Bay et al. 2008; DeTone et al. 2018; Edstedt et al. 2024; Evangelidis and Psarakis 2008; Lindenberger et al. 2023; Lowe 2004; Rublee et al. 2011; Sarlin et al. 2020; Sun et al. 2021]. Robust estimators (LO-RANSAC, USAC, MAGSAC) are essential under modality gaps and outlier-heavy correspondences [Barath et al. 2019; Chum et al. 2003; Raghunandan et al. 2013].

Our raw-scene QA protocol follows this systems view: it reports match quality and geometric stability separately, preserves legacy strict criteria for strong baselines, and uses a selective warning path only for weak-baseline/high-modality-gap cases. This is a protocol contribution rather than a new matcher.

2.4 Depth and Geometry Diagnostics

Depth-based reconstruction metrics are widely used in multi-view stereo and neural reconstruction [Gu et al. 2020; Wang et al. 2021; Yao et al. 2018]. However, raw UAV multispectral captures usually lack globally calibrated metric scale. We therefore evaluate geometry by relative reference-depth agreement on held-out views, and report thresholded agreement curves rather than absolute-meter accuracy claims. This positions our geometry analysis as repeatability/self-consistency under controlled probes, aligned with multispectral product reliability objectives.

3 Method

3.1 Task Definition

Given RGB and multispectral captures (G/R/RE/NIR), we seek a representation that supports both novel-view rendering and stable spectral product synthesis. Let $\mathcal{G}$ denote Gaussian support parameters (position, scale, rotation, opacity, identity) and $\mathcal{A}_b$ denote appearance parameters for band b. The key design choice in this work is to optimize $\mathcal{G}$ once from RGB, then hold $\mathcal{G}$ fixed while learning $\mathcal{A}_b$.

3.2 Stage 1: RGB Geometry Anchor

We first train an RGB Gaussian model with the standard 3DGS optimization pipeline [Kerbl et al. 2023]. RGB is used as the anchor because it provides the highest texture and most stable feature matching in our UAV captures. The resulting checkpoint defines the shared support $\mathcal{G}^\star$ used by all spectral bands.

3.3 Stage 2: Geometry-Locked Spectral Transfer

UMGS-I (independent transfer). For each band $b \in \{G, R, RE, NIR\}$, we initialize from $\mathcal{G}^\star$, freeze all support parameters, and optimize only $\mathcal{A}_b$. This yields four render-compatible spectral models sharing identical Gaussian topology by construction.

UMGS-J (joint band-bank transfer). We keep the same frozen $\mathcal{G}^\star$ but optimize four appearance banks in one joint training state. Training alternates active bands in a deterministic round-robin schedule, updates only the active bank, and writes one authoritative multispectral checkpoint. Per-band models are exported only for compatibility with existing render/metric scripts.

Representation invariants. For both UMGS-I and UMGS-J we enforce and audit:

- fixed Gaussian count and identity;
- fixed xyz/scale/rotation/opacity;
- appearance-only parameter updates in stage 2.

These invariants are the basis for index-stable shared support. A variant that optimizes per-band xyz, scale, rotation, or opacity is therefore not treated as a valid multispectral product model in our protocol, even if it improves single-band rendering. Such a model can still render each band independently, but the i-th Gaussian no longer denotes the same physical support across G/R/RE/NIR, so model-level ratios such as NDVI are no longer defined on a consistent 3D support.

3.4 Raw-Scene Rectification and Baseline-Aware QA

Raw UAV scenes require spectral-to-RGB rectification before Stage 1. We estimate per-band homographies using deterministic candidate sampling and robust aggregation from feature matches. QA is baseline-aware:

- **strict pass**: legacy edge+gradient improvement and geometry checks;
- **pass-with-warning**: allowed only for weak-baseline/high-modality-gap cases that satisfy match quality, stability, gradient gain, edge floor, and outlier-ratio constraints;
- **fail**: unstable geometry or insufficient matching evidence.

Estimator QA is an online gate; final standalone QA on exported rectified outputs is the authoritative decision.

3.5 Products and Geometry Diagnostics

We render G/R/RE/NIR and compute NDVI, GNDVI, and NDRE under a common valid mask. Geometry is evaluated by relative depth agreement on held-out probe views:

$$\delta_{rel} = \frac{D_{model} - D_{ref}}{D_{ref}}.$$

Because global metric scale is not guaranteed in raw UAV reconstructions, we report relative-depth AbsRel and threshold agreements (1%, 5%, 10%) instead of meter-based absolute accuracy.

4 Experiments

4.1 Setup

Scenes. We evaluate the main paper on a 10-scene UAV-focused benchmark: 7 raw UAV scenes (`raw_self`, `raw001–raw006`) and 3 aligned aerial multispectral scenes (`ms_golf`, `ms_lake`, `ms_solar`). Raw scenes test rectification, QA, reconstruction, products, and geometry diagnostics. Aligned scenes remove rectification confounds and are used for external common-mask comparison.

Methods. We report two UMGS variants: UMGS-I, an independent per-band transfer pipeline, and UMGS-J, a joint band-bank pipeline with one authoritative multispectral checkpoint. MS-Splatting is the external baseline on aligned UAV-style scenes. Scratch-MS and Unfrozen-support are representative single-band upper-bound ablations; they are not promoted as product models because they do not preserve the shared Gaussian support required by model-level multispectral fusion.

Metrics. Image quality: PSNR/SSIM/LPIPS. Spectral means average G/R/RE/NIR. Index fidelity: MAE/RMSE/PSNR/SSIM for NDVI, GNDVI, NDRE. Geometry: relative-depth AbsRel and agreement at 1%, 5%, 10%. Cost: non-invasive GPU traces and exported Gaussian counts. Aligned external comparison uses a common-mask protocol; UMGS-I-shim and UMGS-J-shim are resolution-aligned comparison views only, not distinct trained variants.

4.2 Main Results on UAV Scenes

Table 1 shows that UMGS-I and UMGS-J remain near-identical on aggregate quality and geometry. On the 10-scene UAV-focused suite, both reach 0.843 spectral SSIM and 0.844 index SSIM; UMGS-I has a small edge on index RMSE (0.075 vs. 0.077). Depth aggregates are identical because both variants share the same locked support.

This result supports a stable interpretation: UMGS-I remains the conservative primary line, while UMGS-J is a valid joint-checkpoint branch with numerically comparable performance.

4.3 Index-Focused Analysis

Table 2 reports NDVI/GNDVI/NDRE separately. Differences between UMGS-I and UMGS-J are small on NDVI and GNDVI, with a slightly larger NDRE gap favoring UMGS-I in RMSE/PSNR. The joint-bank design is therefore viable but not consistently superior.

Table 1. Main UMGS results over the 10-scene UAV-focused suite. RGB metrics for UMGS-J use the shared UMGS-I RGB anchor. Spectral metrics average G/R/RE/NIR. Index metrics average NDVI/GNDVI/NDRE. Depth uses the relative reference-depth protocol and averages G/R/RE/NIR.

Method	Split	RGB PSNR	RGB SSIM	MS PSNR	MS SSIM	MS LPIPS	Idx. RMSE	Idx. SSIM	AbsRel	Ag@5%	Ag@10%
UMGS-I	Raw 7	24.39	0.840	27.40	0.850	0.136	0.065	0.869	0.019	0.916	0.988
UMGS-I	Aligned 3	25.48	0.810	26.50	0.828	0.230	0.098	0.787	0.053	0.804	0.833
UMGS-I	UAV 10	24.72	0.831	27.13	0.843	0.164	0.075	0.844	0.029	0.883	0.942
UMGS-J	Raw 7	24.39	0.840	27.37	0.850	0.136	0.067	0.869	0.019	0.916	0.988
UMGS-J	Aligned 3	25.48	0.810	26.50	0.828	0.230	0.098	0.787	0.053	0.804	0.833
UMGS-J	UAV 10	24.72	0.831	27.11	0.843	0.164	0.077	0.844	0.029	0.883	0.942

Table 2. UAV-focused 10-scene spectral-index fidelity by vegetation index.

Method	Index	MAE	RMSE	PSNR	SSIM
UMGS-I	NDVI	0.058	0.083	28.49	0.847
UMGS-I	GNDVI	0.059	0.083	28.65	0.823
UMGS-I	NDRE	0.041	0.059	31.53	0.864
UMGS-J	NDVI	0.058	0.083	28.49	0.847
UMGS-J	GNDVI	0.059	0.083	28.66	0.823
UMGS-J	NDRE	0.043	0.063	31.31	0.863

Table 3. Aligned UAV-style common-mask comparison against MS-Splatting. Lower is better for LPIPS, 4-band RMSE, and SAM. UMGS shims are resolution-aligned comparison views, not separate training variants.

Method	PSNR	SSIM	LPIPS	4-band RMSE	SAM°
UMGS-I-shim	27.01	0.853	0.215	0.0545	5.38
UMGS-J-shim	27.01	0.853	0.215	0.0545	5.38
MS-Splatting	27.04	0.853	0.244	0.0541	5.38

4.4 Aligned External Comparison

Table 3 summarizes the aligned UAV-style common-mask comparison against MS-Splatting. The average is deliberately reported as scene-dependent: UMGS-J-shim is lower in LPIPS and has the same mean SSIM/SAM, while MS-Splatting is slightly higher in PSNR and slightly lower in spectral RMSE. Scene-level behavior in Table 4 is mixed, so we do not claim universal dominance over the external baseline.

For raw unaligned data, Table 5 is an applicability diagnostic under the external retained-subset path. It is not the main raw benchmark because it does not represent the full raw protocol used by UMGS-I/UMGS-J.

4.5 Representative Ablations

Table 6 compares the locked UMGS variants with two upper-bound ablations on `raw_self`, `raw002`, and `ms_lake`. The unfrozen branch improves selected image metrics, but it is not a valid product model: on `ms_lake`, product construction fails the shared-support audit with an xyz mismatch of 6.84×10^{-2}. Scratch-MS diverges more strongly, including Gaussian-count mismatch across bands. These failures occur at the exact stage where multispectral products are assembled, showing that per-band quality gains alone are insufficient if support consistency is lost.

4.6 Cost

Table 7 reports coarse but traceable cost indicators on the UAV-focused runs. UMGS-J uses higher recorded peak VRAM but lower average recorded VRAM in the aggregated traces; both UMGS variants retain identical Gaussian counts.

5 Limitations

First, aligned external comparison is scene-dependent rather than one-sided. Some entries favor our geometry-locked shims, while others favor MS-Splatting. Our claim is therefore reliability and protocol applicability, not universal metric dominance.

Second, UMGS-I and UMGS-J are very close numerically. This validates that the geometry-lock principle is robust to independent and joint implementations. In the current paper, UMGS-J is best interpreted as a strong alternative branch rather than a strict replacement for UMGS-I.

Third, our geometry evaluation is a reference-depth consistency protocol, not dense metric ground truth. This is appropriate for cross-band support analysis under missing global scale, but it does not replace LiDAR-grade absolute geometry benchmarks.

Fourth, cost reporting is intentionally non-invasive and currently coarse. The reported VRAM/utilization/size statistics are traceable and reproducible, but a dedicated runtime microbenchmark (e.g., standardized per-view FPS across methods and resolutions) is still future work.

6 Conclusion

We presented UMGS, a geometry-locked multispectral Gaussian reconstruction framework for UAV spectral-index synthesis. The method anchors geometry in RGB, then learns spectral appearance with fixed support, so cross-band products are built on consistent Gaussian topology instead of loosely coupled per-band reconstructions.

Across the UAV-focused evaluation, UMGS-I and UMGS-J achieve near-identical quality and geometry behavior, showing that the geometry-lock principle is stable under both independent and joint spectral parameterizations. On aligned UAV-style scenes, comparison with MS-Splatting is competitive and scene-dependent, while on raw scenes the full pipeline demonstrates practical applicability with rectification QA, product synthesis, and depth-reference diagnostics.

The main takeaway is methodological: for multispectral UAV reconstruction, preserving shared support is a first-order constraint, not a secondary check after optimizing single-band image metrics.

Table 4. Scene-level aligned UAV-style common-mask comparison for UMGS-I-shim and MS-Splatting. PSNR/SSIM/LPIPS average G/R/RE/NIR; 4-band RMSE and SAM evaluate the spectral vector.

Scene	UMGS PSNR	MMS PSNR	UMGS SSIM	MMS SSIM	UMGS LPIPS	MMS LPIPS	UMGS RMSE	MMS RMSE	UMGS SAM	MMS SAM
`ms_golf`	31.33	32.12	0.928	0.935	0.176	0.234	0.0300	0.0273	2.77	3.11
`ms_lake`	22.02	22.79	0.740	0.731	0.350	0.374	0.0887	0.0817	8.84	7.88
`ms_solar`	27.72	26.21	0.893	0.893	0.117	0.123	0.0448	0.0532	4.52	5.16

Table 5. Raw `self_m3m` retained-subset diagnostic. This is an applicability check for the external pipeline under its retained subset, not the main raw full-scene comparison.

Method	PSNR	SSIM	LPIPS	4-band RMSE	SAM°
UMGS-I-retained	28.35	0.834	0.203	0.0502	3.98
UMGS-J-retained	28.33	0.834	0.204	0.0503	4.00
MS-Splatting-default	31.07	0.933	0.107	0.0379	3.95

References

Nicolas Audebert, Bertrand Le Saux, and Sébastien Lefèvre. 2019. Deep Learning for Classification of Hyperspectral Data: A Comparative Review. *IEEE Geoscience and Remote Sensing Magazine* 7, 2 (2019), 159–173.

Dániel Barath, Jiri Matas, and Jana Noskova. 2019. MAGSAC: Marginalizing Sample Consensus. In *CVPR*.

Jonathan T. Barron, Ben Mildenhall, Matthew Tancik, Peter Hedman, Ricardo Martin-Brualla, and Pratul P. Srinivasan. 2022. Mip-NeRF 360: Unbounded Anti-Aliased Neural Radiance Fields. In *CVPR*.

Jonathan T. Barron, Ben Mildenhall, Dor Verbin, Peter Hedman, and Pratul P. Srinivasan. 2023. Zip-NeRF: Anti-Aliased Grid-Based Neural Radiance Fields. *ICCV* (2023).

Jonathan T. Barron, Ben Mildenhall, Dor Verbin, Pratul P. Srinivasan, and Peter Hedman. 2021. Mip-NeRF: A Multiscale Representation for Anti-Aliasing Neural Radiance Fields. In *ICCV*.

Herbert Bay, Andreas Ess, Tinne Tuytelaars, and Luc Van Gool. 2008. Speeded-Up Robust Features (SURF). *Computer Vision and Image Understanding* 110, 3 (2008), 346–359.

Anpei Chen, Zexiang Xu, Andreas Geiger, Jingyi Yu, and Hao Su. 2022. TensoRF: Tensorial Radiance Fields. In *ECCV*.

Qian Chen, Shihao Shu, and Xiangzhi Bai. 2024. Thermal3D-GS: Physics-induced 3D Gaussians for Thermal Infrared Novel-view Synthesis. arXiv:2409.08042 [cs.CV] https://arxiv.org/abs/2409.08042

Ondrej Chum, Jiri Matas, and Josef Kittler. 2003. Locally Optimized RANSAC. In *DAGM*.

Ismael Colomina and Pere Molina. 2014. Unmanned Aerial Systems for Photogrammetry and Remote Sensing: A Review. *ISPRS Journal of Photogrammetry and Remote Sensing* 92 (2014), 79–97.

Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. 2018. SuperPoint: Self-Supervised Interest Point Detection and Description. In *CVPR Workshops*.

Johan Edstedt, Qiyuan Wang, Victor Larsson, Daniel Barath, and Torsten Sattler. 2024. RoMa: Robust Dense Feature Matching. In *CVPR*.

Georgios D. Evangelidis and Emmanouil Z. Psarakis. 2008. Parametric Image Alignment Using Enhanced Correlation Coefficient Maximization. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 30, 10 (2008), 1858–1865.

Zhiwen Fan, Wenqi Xian, Jiaxin Yang, Dejia Xu, Deva Ramanan, Jan Kautz, and Zhe Lin. 2024. LightGaussian: Unbounded 3D Gaussian Compression with 15x Reduction and 200+ FPS. *NeurIPS* (2024).

Martin A. Fischler and Robert C. Bolles. 1981. Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis and Automated Cartography. *Commun. ACM* 24, 6 (1981), 381–395.

Sara Fridovich-Keil, Giacomo Meanti, Francesco Poiesi, Michael Rubinstein, and Angjoo Kanazawa. 2023. K-Planes: Explicit Radiance Fields in Space, Time, and Appearance. In *CVPR*.

Sara Fridovich-Keil, Alex Yu, Matthew Tancik, Qinhong Jiang, Benjamin Recht, and Angjoo Kanazawa. 2022. Plenoxels: Radiance Fields without Neural Networks. In *CVPR*.

Anatoly A. Gitelson, Yosef J. Kaufman, and Mark N. Merzlyak. 1996. Use of a Green Channel in Remote Sensing of Global Vegetation from EOS-MODIS. *Remote Sensing of Environment* 58, 3 (1996), 289–298.

Alexander F. H. Goetz, George Vane, John E. Solomon, and Barry N. Rock. 1985. Imaging Spectrometry for Earth Remote Sensing. *Science* 228, 4704 (1985), 1147–1153.

Xiaodong Gu, Zhiqiang Fan, Siyu Zhu, Zhaoyang Dai, Feitong Tan, and Ping Tan. 2020. CAS-MVSNet: Cascade Cost Volume for High-Resolution Multi-View Stereo and Stereo Matching. In *CVPR*.

Antoine Guédon and Vincent Lepetit. 2024. SuGaR: Surface-Aligned Gaussian Splatting for Efficient 3D Mesh Reconstruction and High-Quality Novel View Synthesis. *CVPR* (2024).

Richard Hartley and Andrew Zisserman. 2004. *Multiple View Geometry in Computer Vision* (2nd ed.). Cambridge University Press.

Binbin Huang, Zehao Yu, Anpei Chen, Andreas Geiger, Shenghua Gao, and Hao Su. 2024. 2D Gaussian Splatting for Geometrically Accurate Radiance Fields. *SIGGRAPH* (2024).

Alfredo R. Huete. 1988. A Soil-Adjusted Vegetation Index (SAVI). *Remote Sensing of Environment* 25, 3 (1988), 295–309.

Bernhard Kerbl, Georgios Kopanas, Thomas Leimk"uhler, and George Drettakis. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics* 42, 4 (2023), 139:1–139:14.

Jaesung Lee, Jihoon Kim, Jaewoo Park, and Minsik Lee. 2024. Compact3DGS: Memory-Efficient 3D Gaussian Splatting. *arXiv preprint arXiv:2403.18159* (2024).

Wen Li, Geng Sun, Yunsong Li, and Qian Du. 2019. Deep Learning for Hyperspectral Image Classification: An Overview. *IEEE Transactions on Geoscience and Remote Sensing* 57, 9 (2019), 6690–6709.

Philipp Lindenberger, Paul-Edouard Sarlin, and Marc Pollefeys. 2023. LightGlue: Local Feature Matching at Light Speed. In *ICCV*.

Lingjie Liu, Jiatao Gu, Kyaw Zaw Lin, Tat-Seng Chua, and Christian Theobalt. 2020. Neural Sparse Voxel Fields. *NeurIPS* (2020).

Yang Liu, He Guan, Chuanchen Luo, Lue Fan, Naiyan Wang, Junran Peng, and Zhaoxiang Zhang. 2024. CityGaussian: Real-Time High-Quality Large-Scale Scene Rendering with Gaussians. *ECCV* (2024).

Yang Liu, Chuanchen Luo, Zhongkai Mao, Junran Peng, and Zhaoxiang Zhang. 2025. CityGaussianV2: Efficient and Geometrically Accurate Reconstruction for Large-Scale Scenes. In *ICLR*.

David G. Lowe. 2004. Distinctive Image Features from Scale-Invariant Keypoints. *International Journal of Computer Vision* 60, 2 (2004), 91–110.

Jiahao Lu, Xiangli Yao, Yinda Zhang, Yao Yao, Hai Li, Tian Fang, and Long Quan. 2024b. Scaffold-GS: Structured 3D Gaussians for View-Adaptive Rendering. *CVPR* (2024).

Rongfeng Lu, Hangyu Chen, Zunjie Zhu, Yuhang Qin, Ming Lu, Le Zhang, Chenggang Yan, and Anke Xue. 2024a. ThermalGaussian: Thermal 3D Gaussian Splatting. arXiv:2409.07200 [cs.CV] https://arxiv.org/abs/2409.07200

Jonathon Luiten, Leonidas Guibas, Deva Ramanan, and Florian Bernard. 2024. Dynamic 3D Gaussians: Tracking by Persistent Dynamic View Synthesis. *3DV* (2024).

Dimitris Manolakis, Ronald Lockwood, Thomas Cooley, and John Jacobson. 2016. *Hyperspectral Imaging Remote Sensing: Physics, Sensors, and Algorithms*. Cambridge University Press.

A. J. Mathews and S. Jensen. 2013. Visualizing and Quantifying Vineyard Canopy LAI Using an Unmanned Aerial Vehicle (UAV) Collected High Density Structure from Motion Point Cloud. *Remote Sensing* 5, 5 (2013), 2164–2183.

Lukas Meyer, Josef Grün, Maximilian Weiherer, Bernhard Egger, Marc Stamminger, and Linus Franke. 2025. Multi-Spectral Gaussian Splatting with Neural Color Representation. arXiv:2506.03407 [cs.CV] https://arxiv.org/abs/2506.03407

Ben Mildenhall, Pratul P. Srinivasan, Rodrigo Ortiz-Cayon, Nima Khademi Kalantari, and Ren Ng. 2019. Local Light Field Fusion: Practical View Synthesis with Prescriptive Sampling Guidelines. In *ACM Transactions on Graphics*.

Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. 2020. NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis. In *ECCV*.

Thomas M"uller, Alex Evans, Christoph Schied, and Alexander Keller. 2022. Instant Neural Graphics Primitives with a Multiresolution Hash Encoding. *ACM Transactions on Graphics* 41, 4 (2022).

M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza. 2019. Deep Learning Classifiers for Hyperspectral Imaging: A Review. *ISPRS Journal of Photogrammetry and Remote Sensing* 158 (2019), 279–317.

Raguram Raghunandan, Ondrej Chum, Marc Pollefeys, Jiri Matas, and Jan-Michael Frahm. 2013. USAC: A Universal Framework for Random Sample Consensus. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 35, 8 (2013), 2022–2038.

Christian Reiser, Songyou Peng, Yuanqing Zhang, and Andreas Geiger. 2023. MERF: Memory-Efficient Radiance Fields for Real-Time View Synthesis in Unbounded Scenes. *SIGGRAPH Asia Conference Papers* (2023).

Table 6. Representative 3-scene ablation summary on `raw_self`, `raw002`, and `ms_lake`. Spectral metrics average G/R/RE/NIR; index metrics average NDVI/GNDVI/NDRE. Scratch-MS and Unfrozen-support are single-band upper-bound ablations, not valid product models, because their shared-support audits fail.

Method	MS PSNR	MS SSIM	MS LPIPS	Idx. RMSE	AbsRel	Ag@5%	Shared	Products
UMGS-I	25.35	0.786	0.228	0.100	0.045	0.833	yes	yes
UMGS-J	25.33	0.786	0.228	0.100	0.045	0.833	yes	yes
Scratch-MS	25.88	0.780	0.289	0.116	0.035	0.560	no	invalid
Unfrozen-support	26.27	0.794	0.218	0.105	0.046	0.830	no	invalid

Table 7. Coarse recorded cost indicators on the UAV-focused suite. Values are aggregated from non-invasive GPU traces and exported PLY vertex counts; they are intended as a first cost table, not a microbenchmark.

Method	Peak VRAM GB	Avg. VRAM GB	Peak util. %	Gaussians M
UMGS-I	27.45	3.99	99.0	4.31
UMGS-J	35.04	1.50	99.0	4.31

John W. Rouse, Robert H. Haas, John A. Schell, and Donald W. Deering. 1974. Monitoring Vegetation Systems in the Great Plains with ERTS. In *Third Earth Resources Technology Satellite Symposium*.

Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski. 2011. ORB: An Efficient Alternative to SIFT or SURF. In *ICCV*.

Paul-Edouard Sarlin, Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. 2020. SuperGlue: Learning Feature Matching with Graph Neural Networks. In *CVPR*.

Johannes L. Sch"onberger and Jan-Michael Frahm. 2016. Structure-from-Motion Revisited. In *CVPR*.

Johannes L. Sch"onberger, Enliang Zheng, Marc Pollefeys, and Jan-Michael Frahm. 2016. Pixelwise View Selection for Unstructured Multi-View Stereo. In *ECCV*.

Cheng Sun, Min Sun, and Hwann-Tzong Chen. 2022. Direct Voxel Grid Optimization: Super-fast Convergence for Radiance Fields Reconstruction. In *CVPR*.

Jiaming Sun, Zixin Shen, Yuangdong Xu, Zerun Wang, Qi Bao, Xiaowei Zhou, Yanning Zhang, and Junsong Yuan. 2021. LoFTR: Detector-Free Local Feature Matching with Transformers. In *CVPR*.

Matthew Tancik, Vincent Casser, Xinchen Yan, Sabeek Pradhan, Ben Mildenhall, Pieter Abbeel, Ren Ng, and Angjoo Kanazawa. 2022. Block-NeRF: Scalable Large Scene Neural View Synthesis. *CVPR* (2022).

Christopher Thirgood, Oscar Mendez, Erin Chao Ling, Jon Storey, and Simon Hadfield. 2025. HyperGS: Hyperspectral 3D Gaussian Splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 5970–5979.

Hadi Turki, Deva Ramanan, and Mahadev Satyanarayanan. 2022. Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs. In *CVPR*.

Darren Turner, Andreas Lucieer, and Simon M. de Jong. 2015. Time Series Analysis of Landslide Dynamics Using an Unmanned Aerial Vehicle (UAV). *Remote Sensing* 7, 2 (2015), 1736–1757.

Fangjinhua Wang, Silvano Galliani, Christoph Vogel, Marc Pollefeys, and Jan Kautz. 2021. PatchmatchNet: Learned Multi-View Patchmatch Stereo. In *CVPR*.

Yao Yao, Zixin Luo, Shiwei Li, Tian Fang, and Long Quan. 2018. MVSNet: Depth Inference for Unstructured Multi-view Stereo. In *ECCV*.

Zehao Yu, Anpei Chen, Andreas Geiger, Shenghua Gao, and Hao Su. 2024. Mip-Splatting: Alias-Free 3D Gaussian Splatting. *CVPR* (2024).

Chao Zhang and John M. Kovacs. 2012. The Application of Small Unmanned Aerial Systems for Precision Agriculture: A Review. *Precision Agriculture* 13, 6 (2012), 693–712.

Fanglue Zhang et al. 2026. MSGS: Multispectral 3D Gaussian Splatting. arXiv:2604.13340 [cs.CV] https://arxiv.org/abs/2604.13340

Zilong Zhong, Jonathan Li, Ziyu Luo, and Michael Chapman. 2018. Spectral-Spatial Residual Network for Hyperspectral Image Classification: A 3-D Deep Learning Framework. *IEEE Transactions on Geoscience and Remote Sensing* 56, 2 (2018), 847–858.